

Ravish Paul

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SUMMARY

AI/LLM-focused B.Tech student with hands-on experience in Retrieval-Augmented Generation (RAG), transformer architectures, prompt injection detection, and LLM fine-tuning using LoRA & PEFT. Skilled in building AI-powered applications using PyTorch, Hugging Face, FastAPI, and Streamlit. Passionate about Machine Learning, NLP, and secure AI systems with strong interest in real-world AI engineering and research.

EDUCATION

Parul University

Bachelor of Technology (B.Tech)

Vadodara, Gujarat

Sept. 2023 – May 2027

M B D College

Major: Physics, Chemistry, Mathematics (PCM) — 63.00% (BSEB)

Darbhanga, Bihar

June 2021 – April 2023

TECHNICAL SKILLS

Programming Languages: Python, Basic SQL

AI / Machine Learning: Machine Learning, Deep Learning, Neural Networks, CNN, RNN, Natural Language Processing (NLP)

LLM & NLP: Transformers, Attention Mechanism, Tokenization, Prompt Engineering, LoRA, PEFT, Retrieval-Augmented Generation (RAG)

Frameworks & Libraries: PyTorch, Hugging Face, Scikit-learn, FastAPI, Streamlit, Pandas, NumPy

Cloud & Tools: AWS, Azure, Git, GitHub, Jupyter Notebook

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOPs), DBMS, Computer Networks, Operating Systems

PROJECTS

Retrieval-Augmented Generation (RAG) Question Answering System

Jun 2025 – Aug 2025

Python, FastAPI, Streamlit, Vector Embeddings

- Developed a Retrieval-Augmented Generation (RAG) pipeline for document-based question answering.
- Integrated vector embedding-based document retrieval for contextual response generation.
- Built REST APIs using FastAPI and interactive UI using Streamlit.
- Improved response relevance through optimized retrieval and embedding workflows.

LLM Fine-Tuning using LoRA & PEFT

Aug 2025 – Oct 2025

PyTorch, Hugging Face, Transformers

- Fine-tuned pre-trained Large Language Models on custom datasets using LoRA and PEFT techniques.
- Reduced computational requirements while improving task-specific model performance.
- Worked on tokenizer optimization, prompt formatting, and transformer-based workflows.
- Evaluated model performance using training and validation loss metrics.

Transformer Architecture from Scratch

Oct 2025 – Dec 2025

PyTorch, Deep Learning, NLP

- Implemented encoder-decoder transformer architecture from scratch using PyTorch.
- Developed self-attention and positional encoding mechanisms for sequence learning tasks.
- Trained and tested the model on text datasets for NLP-based applications.
- Explored attention visualization and optimization techniques for better model understanding.

PromptArmor – AI Firewall for Prompt Injection Detection

Jan 2026 – Present

Python, FastAPI, Transformers, NLP, Cybersecurity

- Developed an AI security system for detecting and blocking prompt injection attacks in LLM applications.
- Built real-time threat detection pipelines for identifying jailbreak attempts, malicious prompts, and hidden instruction attacks.

- Implemented rule-based and NLP-based filtering techniques for prompt risk analysis and classification.
- Designed a risk-scoring mechanism to allow, sanitize, or block suspicious prompts before reaching the LLM.
- Worked on secure AI workflow architecture for protecting AI chatbots and LLM-based systems.

ACHIEVEMENTS

- **AWS Cloud Certification:** Completed cloud computing training covering cloud concepts, core AWS services, deployment, and security fundamentals.
- **NPTEL Certification:** Computer Networks: Successfully completed NPTEL certification focused on networking concepts, routing, protocols, and communication systems.
- Solved more than **100+ coding problems** across various programming platforms.
- Built AI/LLM projects including RAG pipelines, transformer architectures, and LLM fine-tuning systems.